

Met Office plant pest, pathogen & biosecurity research

Relevance to UK climate – nature policy activities · Assessment · Prepared June 2026

Key points

- The Met Office runs the Defra-funded **Climate and Plant Biosecurity Climate Service**. Its degree-day **Climate-Pest Risk web tool** and forest-microclimate work feed Defra / APHA surveillance, contingency planning and the GB Plant Health Risk Register — the clearest, documented research-to-policy route.
- **Currently informed:** Plant Biosecurity Strategy for GB 2023–28, the GB Plant Health Risk Register and contingency / outbreak response, the Tree Health Resilience Strategy, the tree-health component of the England Trees Action Plan, and CCRA3 — all already draw on MO climate-pest outputs.
- **Opportunities to inform (not yet realised):** CCRA4 / NAP4 (2026–27, the highest-value entry point); EIP 2025 and NBSAP 2025 invasive-species commitments; NCEA woodland-condition attribution; ELM, Woodland Carbon Code permanence and Land Use Framework siting; routing pest science through UKNCSP / the NAP Biodiversity & Ecosystems Working Group; and targeting Observatree surveillance with MO climate-suitability maps.
- **Six policy indicators** are clearly informed (or informable) by MO work — invasive non-native species distribution and establishment (EIF, CCC, JNCC), crop / livestock pest and disease incidence (CCC), and the Spring Index phenology indicator (JNCC). See table before Methods.
- **Caveats:** the governance workbook holds no records for the risk register, contingency plans or Observatree; the carbon / biomass / land-use links are credible but not yet documented.

Summary of relevant Met Office research

The relevant Met Office (MO) work clusters around the **Vegetation–Climate Interactions team** and the Defra-funded **Climate and Plant Biosecurity Climate Service** (MO Hadley Centre with Defra’s Plant Health Risk and Horizon Scanning team, University of Exeter, Fera Science, University of Warwick, Forest Research and Kew). Its core outputs are summarised below.

- **UK Climate-Pest Risk web tool** — maps accumulated day-degrees above a base temperature (default 10°C) to indicate where and when climate is suitable for pest development and emergence across the UK, for predefined quarantine pests (Oak Processionary Moth, Asian longhorn beetle, *Popillia japonica*, Colorado potato beetle, brown marmorated stink bug) plus user-defined thresholds.
- **Microclimate model development and validation** — a February 2024 campaign (MO, Exeter, Forest Research, Kew) installed sensors monitoring temperature within woodlands and inside tree trunks at five southern-England sites to improve modelling of pest/pathogen microclimates and biosecurity risk estimates.
- **Pest phenology and establishment modelling** — peach-potato aphid early-flight risk (Hemming et al. 2022), *Bemisia tabaci* UK-establishment limits under climate change (Bradshaw et al. 2019), wheat stem-rust transmission pathways (Bradshaw et al. 2024), Fall Armyworm climate suitability (Urhausen et al. 2024).
- **Foundational review** — Hemming & Macneill (2020), *Use of meteorological data in biosecurity*, sets out how met and climate data feed pest risk, surveillance and response.

Defra partners with the Met Office to provide predicted pest emergence dates to aid outbreak responses — the clearest research-to-policy channel.

Policy activities, relevance scores and rationale

Two independent axes. **Relevance** (materiality of MO science to the policy): **3 High** = a direct, often named evidence input; **2 Medium** = informs a defined risk or evidence component; **1 Low** = indirect dependency — shown by row colour. **Status:** Current = MO science already informs the activity (documented); Opportunity = a credible but not-yet-realised use. Items marked † are activities not yet held as records in the workbook. Policy names are clickable links to the source page.

Ref	Policy activity	Relevance	Status	Why / how it is (or could be) informed by MO research
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Ref	Policy activity	Relevance	Status	Why / how it is (or could be) informed by MO research
POL-034	Plant Biosecurity Strategy for Great Britain 2023–28	3 HIGH	Current	The strategy's prevention, preparedness, response and surveillance outcomes are the direct user of the climate-pest tool and Climate Service. MO emergence-date forecasts and climate-suitability maps already feed contingency planning and surveillance targeting under this strategy (documented partnership).
POL-033	Tree Health Resilience Strategy 2018	3 HIGH	Current	Aimed squarely at tree pest/disease threats (OPM, <i>Xylella</i> , ash dieback); the forest microclimate monitoring campaign and tree-trunk thermal modelling were designed around exactly these risks. MO outputs inform risk-based surveillance and the resilience goals (condition, diversity).
POL-011	England Trees Action Plan 2021–24 (Tree Health Policy component)	3 HIGH	Current	Woodland-creation targets depend on establishing pest- and disease-resilient stock; the embedded Tree Health Policy and Observatree surveillance draw on MO climate-pest risk mapping for trees. Establishment failure from pests directly threatens the 7,000 ha/year target.
†	GB Plant Health Risk Register / Defra Plant Health Risk & Horizon Scanning	3 HIGH	Current	The operational register rating pest likelihood and impact; climate-suitability and degree-day outputs feed the likelihood/establishment components for temperature-limited pests. This is the day-to-day decision tool the MO service was built to support.
†	Defra generic & pest-specific contingency plans / GB Plant Health Service outbreak response	3 HIGH	Current	Defra partners with the Met Office to provide predicted pest emergence dates to aid outbreak responses. Emergence timing drives inspection scheduling and treatment windows during eradication.
POL-017	Climate Change Risk Assessment (CCRA3 2022)	3 HIGH	Current	Pests, pathogens and INNS are explicit UK climate risks and CCRA3 already drew on MO climate evidence. Opportunity: CCRA4 (2026–27) is a near-term, high-value insertion point to expand the climate-pest projections feeding the next assessment and NAP4.
POL-002	National Adaptation Programme (NAP3) 2023–28	2 MED	Opportunity	Shifting pest/pathogen ranges are a land-and-nature adaptation risk; MO climate-suitability modelling could inform how surveillance and tree-health measures adapt, linking CCRA evidence to deliverable actions. Not yet a defined input.
POL-001	Environmental Improvement Plan 2025	2 MED	Opportunity	Woodland and habitat condition targets and INNS commitments under the Environment Act 2021 are sensitive to pest/pathogen pressure; MO research could inform the biosecurity-relevant elements and the relevant EIF condition indicators.
POL-003	UK National Biodiversity Strategy and Action Plan 2025	2 MED	Opportunity	Delivers GBF Target 6 (invasive non-native species); MO climate-suitability tools could help anticipate which INNS become establishment-capable under UK warming.
POL-036	NAP Biodiversity & Ecosystems Working Group	2 MED	Opportunity	Advises on biodiversity/ecosystem adaptation, where pest-driven habitat condition change is a climate–nature interaction MO research quantifies. A natural, not-yet-used translation route for the science into NAP reporting.
POL-025	UK National Climate Science Partnership	2 MED	Opportunity	Met Office-led; the coordinating mechanism for translating climate data into CCRA/NAP and policy. The pest-science-specific channel is latent rather than active — an enabler to route climate-pest outputs into policy.
POL-026	Natural Capital & Ecosystem Assessment Programme	2 MED	Opportunity	Its ecosystem-condition baselines intersect with tree-health/pest impacts on woodland condition; interoperable MO and NCEA data could jointly attribute condition change.
POL-009	Land Use Framework 2025	1 LOW	Opportunity	Spatial woodland-expansion decisions could screen for climate-pest risk, but this is not yet an articulated function.
POL-013	Environmental Land Management schemes	1 LOW	Opportunity	SFI / Countryside Stewardship tree-health and biosecurity actions could draw on MO risk mapping for pest-resilient establishment; linkage currently indirect.

Ref	Policy activity	Relevance	Status	Why / how it is (or could be) informed by MO research
POL-040 / POL-030	Nature for Climate Fund / Woodland Carbon Code	1 LOW	Opportunity	Carbon permanence (100-year) is threatened by pest/pathogen loss; MO tree-health risk evidence is relevant to additionality/permanence risk but not yet integrated.
POL-018 / POL-048	Seventh Carbon Budget / Biomass Strategy	1 LOW	Opportunity	Forestry sink and biomass-feedstock assumptions implicitly depend on tree health under climate change; relevance is real but distal.
POL-027	UK Environmental Observation Framework	1 LOW	Opportunity	Coordinates monitoring; could reduce duplication between MO / Forest Research surveillance and other observation networks.
†	Observatree (citizen-science tree-health surveillance)	2 MED	Opportunity	Survey effort could be targeted using MO climate-suitability maps to focus volunteers where establishment risk is rising.

Policy indicators

A scan of *biodiversity_indicators_manual_v6* across its framework sheets for invasive-species, plant-pest/pathogen, tree-health and phenology indicators. Six map clearly onto MO outputs — the climate-pest distribution/establishment modelling, crop-pest phenology, and the Spring Index (which MO produces via the *State of the UK Climate* phenology supplement). Policy applications link to source pages.

Indicator (ref)	Indicator name	Source framework	Current policy application(s)
IND-E-053	Distribution of invasive non-native species and plant pests and diseases	Environmental Indicator Framework (EIF)	EIP 2025 (Environment Act 2021); GBF Target 6
IND-E-052	Abatement of the number of invasive non-native species entering and establishing	Environmental Indicator Framework (EIF)	EIP 2025 (Environment Act 2021); GBF Target 6
IND-C-107	Invasive non-native species – number of new establishments (UK)	CCC Adaptation Monitoring Framework 2023–25	NAP3 / CCRA (Climate Change Act 2008, s.58)
IND-C-113	Incidence of crop and livestock pests and diseases (UK)	CCC Adaptation Monitoring Framework 2023–25	NAP3 / CCRA (Climate Change Act 2008, s.58)
IND-J-042	Pressure from invasive non-native species – establishment rate	JNCC UK Biodiversity Indicators 2025	NBSAP 2025 ; Environment Act 2021 species targets
IND-J-063	Spring Index (phenology)	JNCC UK Biodiversity Indicators 2025	NBSAP 2025 ; Environment Act 2021 species targets

Methods

- MO research scoped** from the supplied bibliography (Hemming ResearchGate profile, 127 publications), filtering to plant pest/pathogen/biosecurity and adjacent vector work; the tool was characterised by inspecting the web app and its specific-pests reference list (PDF), confirming it as a degree-day thermal-accumulation model.
- Partnership and research-to-policy channels** confirmed via the Met Office blog (21 Mar 2024) and Defra's UK Plant Health Information Portal (contingency-planning and strategy pages, accessed June 2026), plus GOV.UK APHA surveillance guidance.
- Policy candidates** drawn from the governance workbook's Policies & Activities sheet ([uk_climate_nature_governance](#); the POL records scored are stable across v8–v15), filtered first on the Biosecurity / Nature / Land sector tags, then all records screened for a plausible pest/pathogen/tree-health dependency; supplemented with operational activities (risk register, contingency plans, Observatree) absent from the workbook.

4. **Scoring** used two independent axes. *Relevance* (materiality of MO pest science to the policy): direct/named input = 3; defined risk-or-adaptation evidence role = 2; implicit dependency = 1. *Status*: Current where MO science already informs the activity (documented), Opportunity where the use is credible but not yet realised. Kept separate so a high-materiality policy not yet using the science (e.g. CCRA4, EIP 2025) is not mistaken for a low-relevance one.
5. **Indicator scan** of biodiversity_indicators_manual_v6 across its framework sheets, keyword-filtered for invasive/non-native species, plant pests and pathogens, tree health and phenology, then narrowed to indicators whose measured quantity MO outputs could directly inform (distribution/establishment modelling, crop-pest phenology, the temperature-driven Spring Index).

Main gaps and caveats

The workbook holds no records for the GB Plant Health Risk Register, contingency plans or Observatree — these are the highest-relevance operational activities and are arguably worth adding as POL records. Relevance for the carbon / biomass / land-use policies is inference, not evidenced in current policy text. The tool covers a small predefined pest set; pathogen (versus insect) coverage is thinner, so pathogen-heavy policies (e.g. *Phytophthora*, *Xylella*) are served more by the microclimate work than by the public tool.

Additional references used

Met Office research (from bibliography)

6. Hemming, D.L. & Macneill, K. (2020) Use of meteorological data in biosecurity. *Emerging Topics in Life Sciences* 4(5).
7. Mosedale, J., Eyre, D., Korycinska, A., Maclean, I.M.D. et al. (2024) Mechanistic microclimate models and plant pest risk modelling. *Journal of Pest Science*.
8. Bradshaw, C., Hemming, D.L., Baker, R., Korycinska, A. et al. (2019) A novel approach for exploring climatic factors limiting current pest distributions: *Bemisia tabaci* in north-west Europe and potential future UK establishment under climate change. *PLOS ONE*.
9. Hemming, D.L., Bell, J., Collier, R., Scaife, A.A. et al. (2022) Likelihood of extreme early flight of *Myzus persicae* across the UK. *Journal of Economic Entomology*.
10. Bradshaw, C., Hemming, D.L., Mona, T., Gilligan, C.A. et al. (2024) Transmission pathways for the stem rust pathogen into Central and East Asia and the role of the alternate host, barberry. *Environmental Research Letters*.
11. Urhausen, S., Bradshaw, C., Davie, J., Zhang, F. et al. (2024) Climate-related risk to maize crops in China from Fall Armyworm, *Spodoptera frugiperda*. *Journal of Pest Science*.
12. Crowhurst, R. & Hemming, D.L. (2022) Forecasting is not just for the weather (peach-potato aphid risk). *The Vegetable Farmer*, April 2022.

Other sources

- Met Office UK Climate-Pest Risk web tool and specific-pests reference list. metoffice.gov.uk/hadobs/biosecurity_uk_hist
- Met Office (21 Mar 2024) Met Office scientists protecting our forests from pests and pathogens (blog).
- Defra, UK Plant Health Information Portal — Contingency planning; Plant Biosecurity Strategy pages (accessed June 2026).
- GOV.UK, APHA plant health surveillance programme.
- uk_climate_nature_governance_v15.xlsx, Policies & Activities sheet (links taken from the Link column).
- biodiversity_indicators_manual_v6.xlsx, framework indicator sheets (EIF, CCC, JNCC).